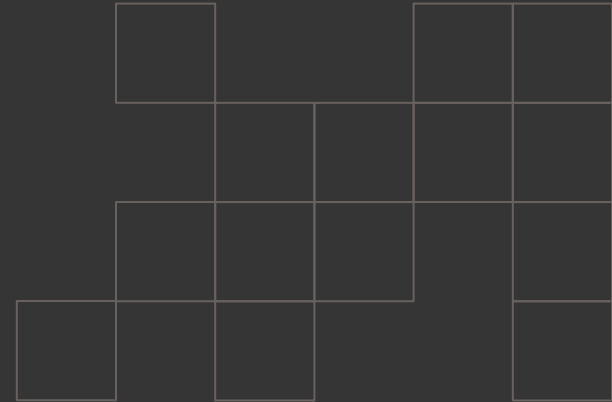


UBS-304 Advanced Transportation Planning

Rebalancing Mobility at Bridletowne Circle

Vivek Khanna



AGENDA

1. CONTEXT

2. NETWORK

3. TRANSIT NETWORK

4. PROBLEM

5. SAFETY

6. INFRASTRUCTURE GAP

7. EQUITY

8. POLICY

9. SOLUTIONS

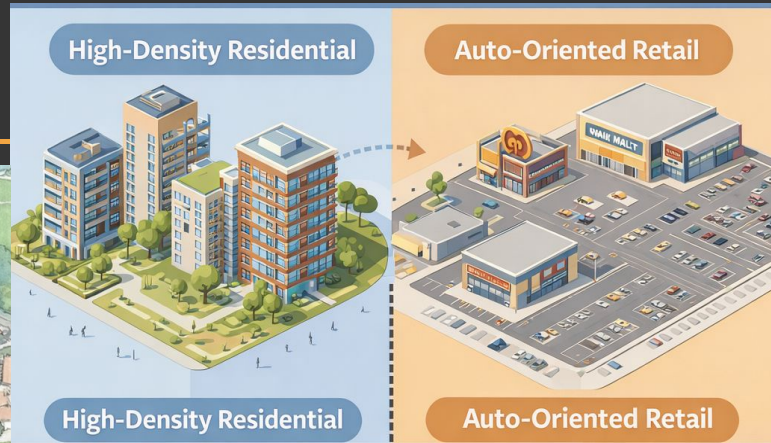
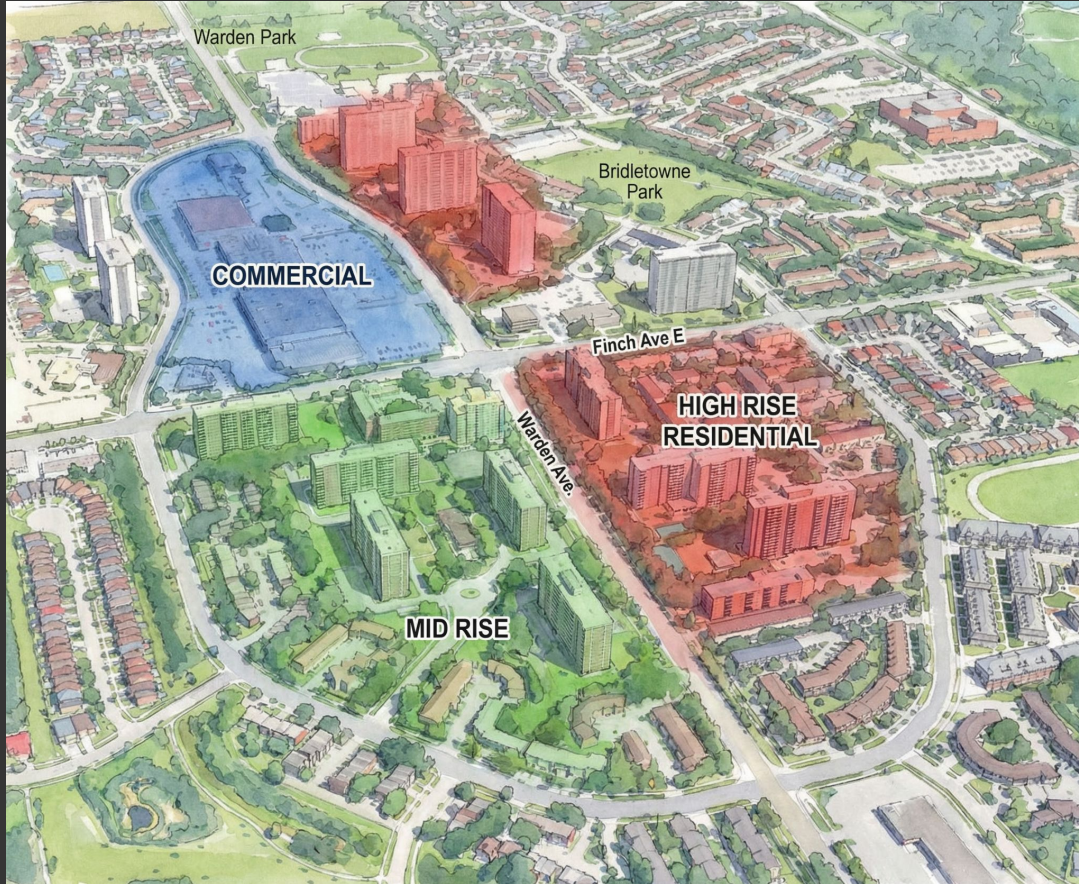
10. CONCLUSION

11. REFERENCES

12. QUESTIONS?

CONTEXT

High-Density Residential Meets Auto-Oriented Retail



Hostile pedestrian environment created by high-speed, 6-lane arterial road crossings.

Surface parking physically separating high-rise residents from essential retail services.

Transit-dependent demographics forced to rely on buses caught in general traffic congestion.

Monofunctional zoning that prevents integrated mixed-use development at the street level.

Absence of a fine-grained street network, making short trips on foot circuitous and difficult.

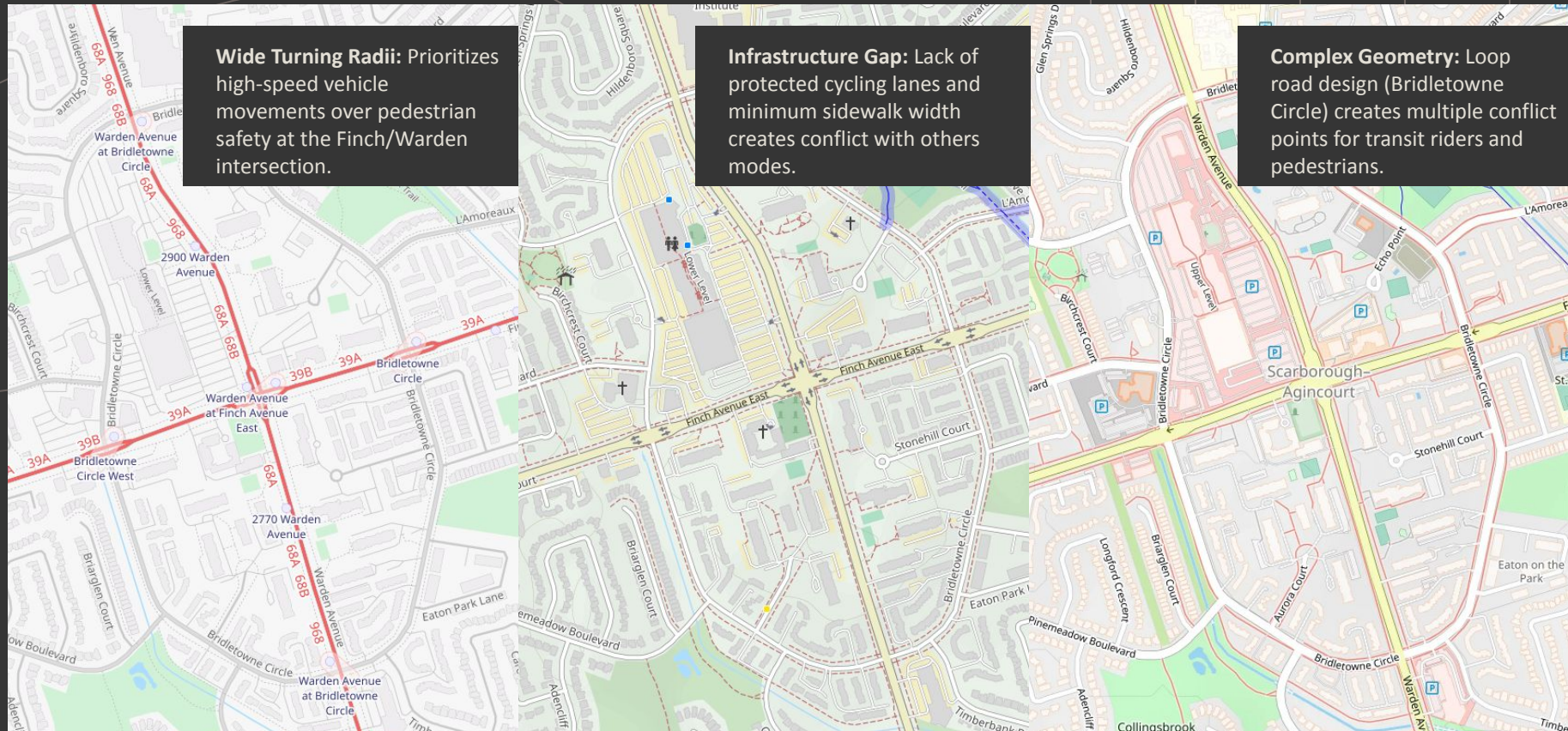
Lack of protected micro-mobility infrastructure (bike lanes) despite high local travel demand.

NETWORK

Wide Turning Radii: Prioritizes high-speed vehicle movements over pedestrian safety at the Finch/Warden intersection.

Infrastructure Gap: Lack of protected cycling lanes and minimum sidewalk width creates conflict with others modes.

Complex Geometry: Loop road design (Bridletowne Circle) creates multiple conflict points for transit riders and pedestrians.



TRANSIT LAYOUT

CYCLE PATH LAYOUT

SIDEWALK & PATHWAY LAYOUT

TRANSIT NETWORK

Finch East Corridor

High-frequency east-west connectivity served by Routes 39A and 39B.

Warden Avenue Axis

Major north-south artery with 68A, 68B, and the 968 Express service.

Lacks Internal Loop Access

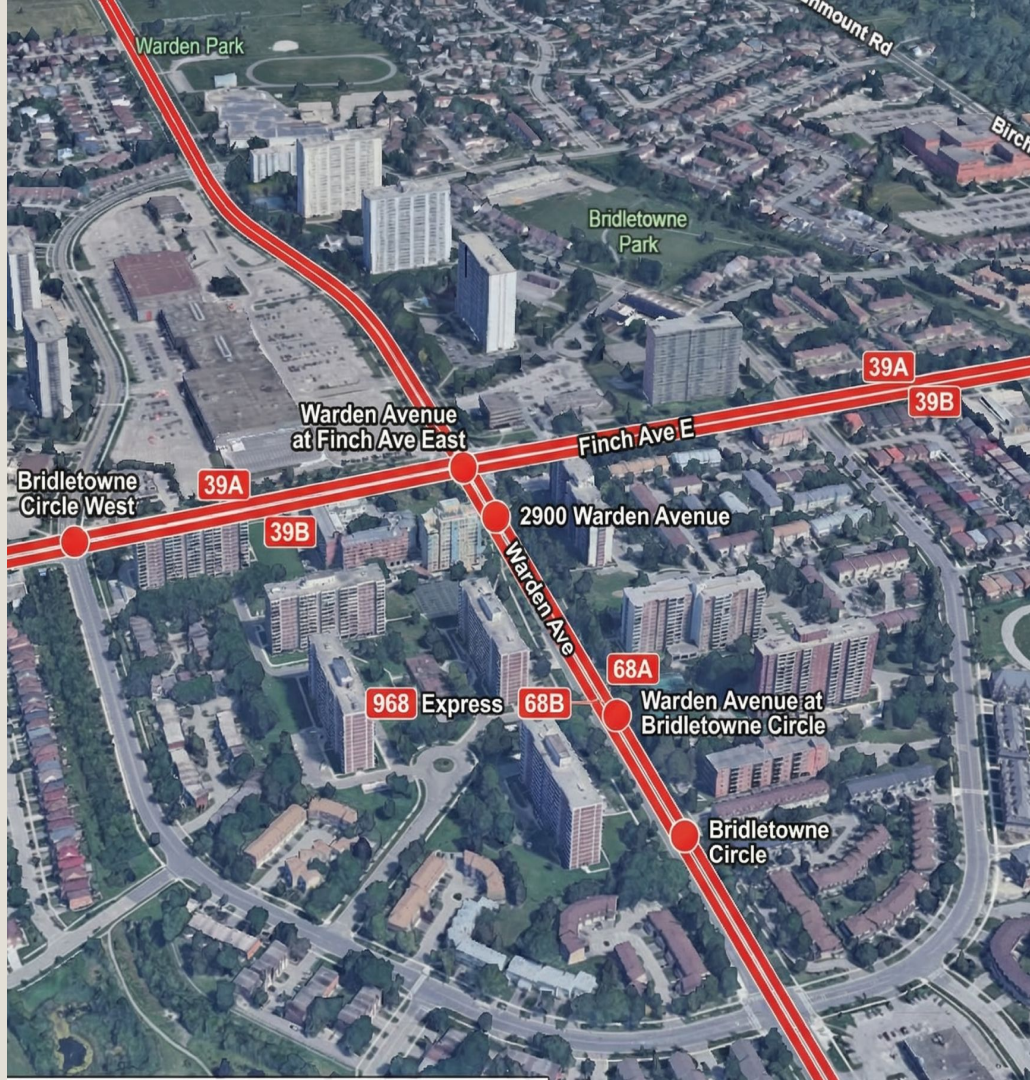
The absence of transit loops or internal circulators within large-scale developments (like Bridlewood Mall or high-rise complexes) creates a last-mile barrier for seniors and those with limited mobility.

Arterial Confinement

Transit service is strictly limited to the peripheral arterial roads, meaning high-capacity vehicles never penetrate the interior of the dense residential clusters.

The Superblock Penalty

Residents in high-density towers are forced into long, exposed walks across vast parking lots and private driveways just to reach the arterial bus stops.



PROBLEM



Bus Stuck in Traffic due to Right turning cars, and lack of Protected Cycling Infrastructure



Lack of Shaded pathway at Major Intersection and Long Unsafe Crossings

Car-centric design undermines the needs of transit-dependent and pedestrian-heavy community.

Unsafe Crossings



Long exposure times and multiple conflict points for pedestrians.

Transit Delays



Unreliable service due to lack of priority measures in mixed traffic.

Bridletowne Circle serves a transit-dependent, pedestrian-heavy residential population but is designed exclusively for high automobile access.

Cycling Gaps



Absent infrastructure creates dangerous network discontinuities.

Excessive Parking



Vast surface lots reduce walkability and waste valuable land.

SAFETY

Multiple Conflict Points

Multiple vehicle turning movements create unpredictable conflict zones.

Lack of adequate refuge islands leaves pedestrians exposed.

High Collision Risk

Seniors and mobility-impaired users face significant barriers, deterring walking trips to local retail.

Exposure Metrics

Crossing Dist ~36.0 m

Walk Speed (Avg) 1.2 m/s

Exposure Time 30.0 sec

Refuge Island None

Turning Vehicle Conflicts

High-speed right turns on red (RTOR) conflict with pedestrians starting their crossing phase.

Wide Turn Radii

Encourages higher turning speeds (30-40 km/h) reducing reaction time for drivers.

Plaza Driveway Chaos

Multiple uncontrolled access points create unpredictable vehicle paths across sidewalks.

Short-Term Fixes

Leading Pedestrian Interval (LPI)

Ban Right-Turn-On-Red (RTOR)

High-visibility zebra markings

Temporary median refuge islands

60 km/h

Posted Speed

6 Lanes

Cross-Section

~35m

Crossing Width

No

Bike Infra

High

Conflict Risk

INFRASTRUCTURE GAP

Network Discontinuity Suppresses Active Travel



Current State: Zero Protection

Bridletowne Circle lacks any protected cycling infrastructure. While Finch Avenue is identified in city plans, the current reality is a high-stress environment that deters riders.

The Intersection Barrier

High-speed traffic and wide turning radii act as a physical wall, severing east-west connectivity for active transportation users.

Missed Opportunity

Dense residential population + short trips to retail = High potential for cycling mode share that is currently suppressed by design.

POLICY

Existing Design Contradicts City Objectives

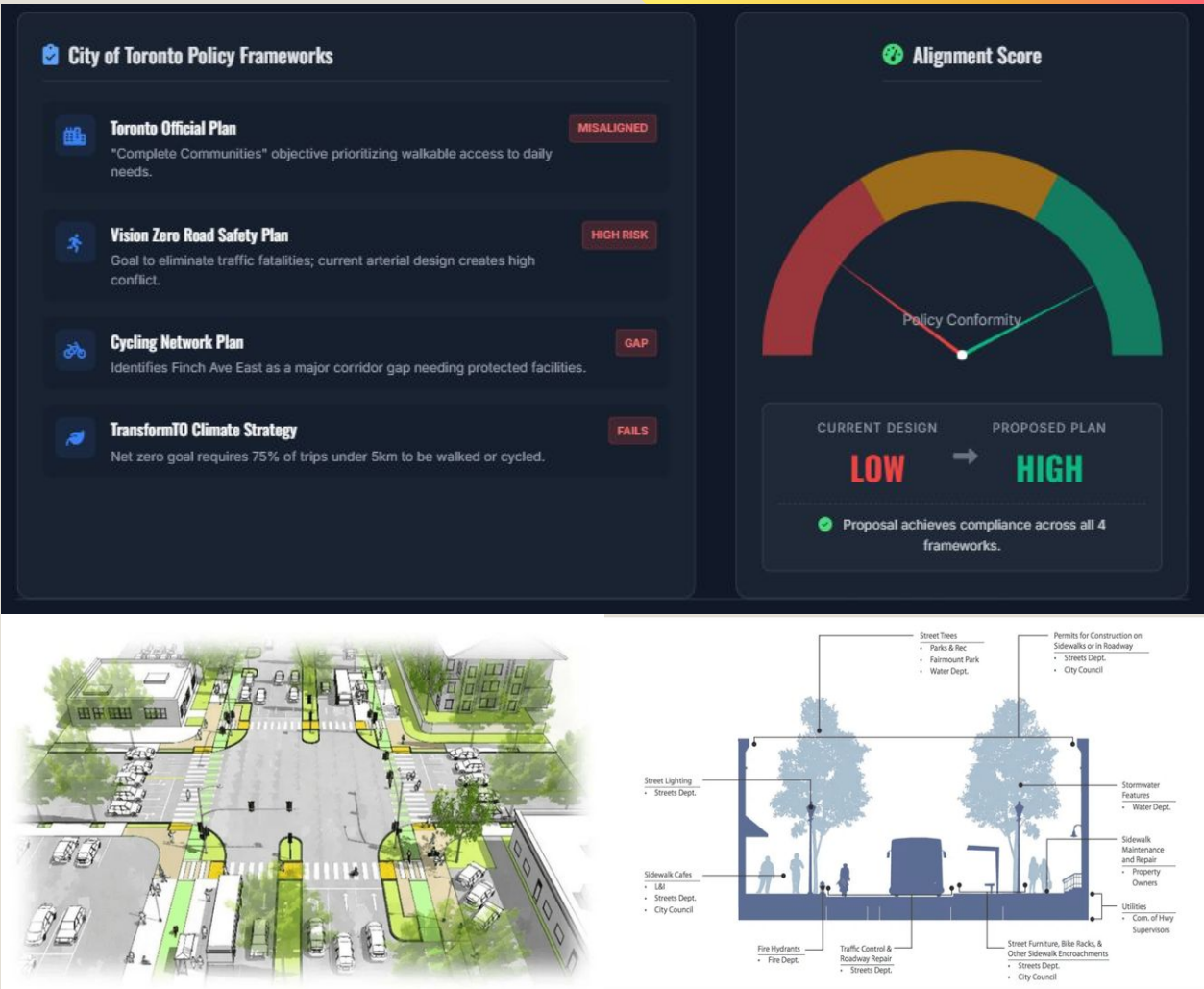
Toronto Official Plan

Vision Zero Road Safety Plan

Complete Streets Guidelines

Toronto Cycling Network Plan

Current infrastructure design at Bridletowne Circle fails to align with stated city policy objectives prioritizing safety, equity, and multimodal access.



SOLUTION

Short-Term



Leading Pedestrian Intervals

Give pedestrians a head start at signals.



Transit Signal Priority

Reduce bus delays at intersections.



Marked Crossings

High-visibility paint at all driveway entrances.

Medium-Term



Curb Extensions

Physically shorten crossing distances.



Protected Cycling Lanes

Fill the network gap on Finch Ave.

Long-Term



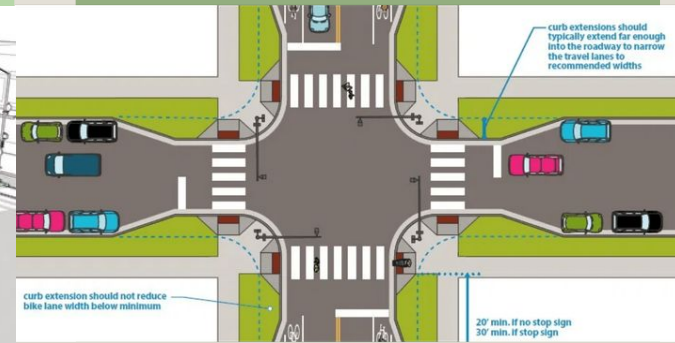
Internal Plaza Redesign

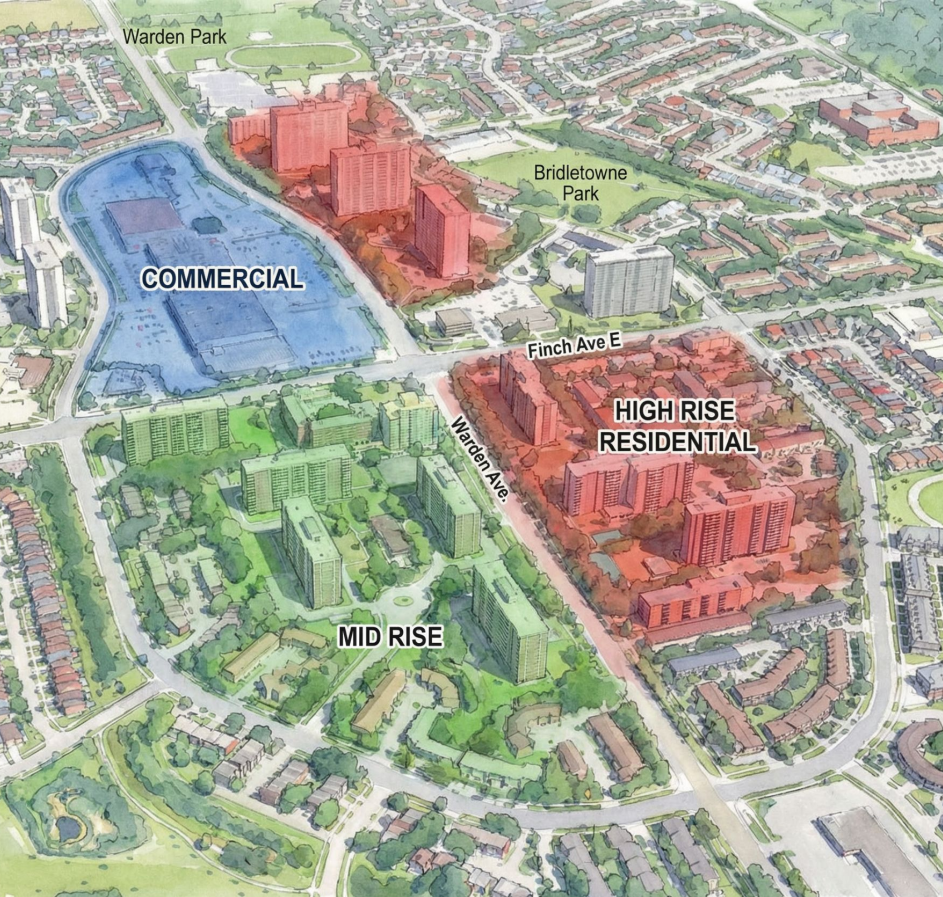
Create a pedestrian-friendly internal grid.



Structured Parking

Replace surface lots to reclaim land.





Different Functions Are Separated Into Distinct Areas.
 Tower in a Park and disconnected walkways
 Lack of ridership adds to the delays and low frequency transit service



Mixed-Use, Fine-Grain Urban Structure
 The Proposed Concept Increases The Amount Of Active Frontage Along
 Primary Streets And Public Spaces
 More Pedestrian Routes And Connections Between Blocks

Suburban Retrofit Challenges Require Multimodal Thinking

Broader Context: Bridletowne Circle exemplifies the challenge of retrofitting post-war suburban infrastructure for modern needs.

The Conflict: Infrastructure designed solely for vehicle efficiency actively harms high-density, transit-oriented communities.

The Solution: Phased interventions can reallocate existing right-of-way to improve safety without massive expansion.

Policy Gap: Success requires translating high-level city goals (Vision Zero) into specific physical design changes.

Should suburban retail nodes
prioritize vehicle efficiency or
multimodal safety and equity
when serving transit-dependent
communities?



SAFER CROSSINGS

Reduced exposure time via refuge islands and LPIs. Zero fatalities target met.



RELIABLE TRANSIT

TSP and queue jumps maintain 10-min headways on Finch East corridor.



PROTECTED CYCLING

Continuous facilities connect local towers to regional network spine.



ACTIVE FRONTAGE

Consolidated driveways create safe pedestrian realm at plaza edge.



MOBILITY HUB

Infrastructure matches the demographics of a transit-dependent community.

REFERENCES

City of Toronto Official Plan

Google Maps & Street View (2025)

Toronto Complete Streets Guidelines

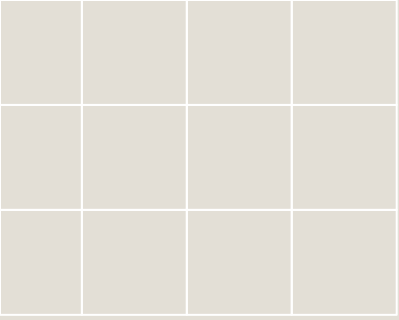
Toronto Cycling Network Plan

City of Toronto Vision Zero Road Safety Plan

City of Toronto Vision Zero Road Safety Plan

TTC Route 39 Finch East

NACTO Urban Street Design Guide



Questions?

